

T3 group factors: UV

Active theory: SM + $S_1 + S_2 + F$. The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$

$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)}(S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2, \ Y_F = (\alpha + 1)/2, \ Y_{S_2} = (\alpha + 2)/2$
$C_2(d)$	$(d^2 - 1)/4$
$T(d)$	$d(d^2 - 1)/12$
$C_{LLS_1S_2}^{ij:AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Field representations

run	S_1	S_2	F	α
T3 – A – m4	(1, −2)	(3, −1)	(2, − $\frac{3}{2}$)	−4
T3 – A – m2	(1, −1)	(3, 0)	(2, − $\frac{1}{2}$)	−2
T3 – A – p0	(1, 0)	(3, 1)	(2, $\frac{1}{2}$)	0
T3 – B – m3	(2, − $\frac{3}{2}$)	(2, − $\frac{1}{2}$)	(1, −1)	−3
T3 – B – m1	(2, − $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(1, 0)	−1
T3 – B – p1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(1, 1)	1
T3 – C – m3	(2, − $\frac{3}{2}$)	(2, − $\frac{1}{2}$)	(3, −1)	−3
T3 – C – m1	(2, − $\frac{1}{2}$)	(2, $\frac{1}{2}$)	(3, 0)	−1
T3 – C – p1	(2, $\frac{1}{2}$)	(2, $\frac{3}{2}$)	(3, 1)	1
T3 – D – m2	(3, −1)	(1, 0)	(2, − $\frac{1}{2}$)	−2
T3 – D – p0	(3, 0)	(1, 1)	(2, $\frac{1}{2}$)	0
T3 – D – p2	(3, 1)	(1, 2)	(2, $\frac{3}{2}$)	2
T3 – E – m4	(3, −2)	(3, −1)	(2, − $\frac{3}{2}$)	−4
T3 – E – m2	(3, −1)	(3, 0)	(2, − $\frac{1}{2}$)	−2
T3 – E – p0	(3, 0)	(3, 1)	(2, $\frac{1}{2}$)	0
T3 – E – p2	(3, 1)	(3, 2)	(2, $\frac{3}{2}$)	2

Gauge and Yukawa beta functions

$$\beta_{g_1}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_1^3	$\frac{163}{6}$	$\frac{55}{6}$	$\frac{59}{6}$	$\frac{25}{2}$	$\frac{43}{6}$

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_1^3	$\frac{25}{2}$	$\frac{41}{2}$	$\frac{43}{6}$	$\frac{41}{2}$	$\frac{59}{6}$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_1^3	$\frac{55}{6}$	$\frac{163}{6}$	$\frac{179}{6}$	$\frac{59}{6}$	$\frac{59}{6}$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_1^3	$\frac{179}{6}$

$$\beta_{g_2}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_2^3	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{17}{6}$	$-\frac{17}{6}$

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_2^3	$-\frac{17}{6}$	$\frac{31}{6}$	$\frac{7}{6}$	$\frac{31}{6}$	$-\frac{1}{2}$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_2^3	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_2^3	$\frac{1}{6}$

$$\beta_{g_3}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_3^3	-7	-7	-7	-7	-7

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_3^3	-7	-7	-7	-7	-7

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
g_3^3	-7	-7	-7	-7	-7

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_3^3	-7

$$\beta_{Y_u}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$

$g_3^2(Y_u)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$

$g_3^2(Y_d)_{ij}$	-8	-8	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta_{Y_e}^{(1)}$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$

$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1	1	1

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(y_1(y_1)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(y_2(y_2)^\dagger Y_e)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$$\beta_{y_1}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$T_\nu^{(1)}(y_1)_{ij}$	2	2	2	1	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	1	1	1	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{15}{4}$	$-\frac{3}{4}$
$\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0	0	1
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0	0	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$T_\nu^{(1)}(y_1)_{ij}$	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-\frac{15}{4}$	$-\frac{3}{2}$
$\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	$\frac{3}{2}$	0	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	$\frac{1}{2}$	0	0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$T_\nu^{(1)}(y_1)_{ij}$	1	1	1	1	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$
$\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0	0	0	0	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0	0	0	0	0

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$T_\nu^{(1)}(y_1)_{ij}$	1
$(y_2(y_2)^\dagger y_1)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_1)_{ij}$	$\frac{1}{2}$
$(y_1(y_1)^\dagger y_1)_{ij}$	$\frac{3}{2}$
$g_2^2(y_1)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_1)_{ij}$	$-\frac{15}{2}$
$\text{Tr}(y_1(y_2)^\dagger)(y_2)_{ij}$	0
$(y_1(y_2)^\dagger y_2)_{ij}$	0

$$\beta_{y_2}^{(1)}$$

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$T_\nu^{(2)}(y_2)_{ij}$	1	1	1	1	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0	0	1
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0	0	1

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$T_\nu^{(2)}(y_2)_{ij}$	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	2

$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$	$\frac{5}{4}$	$\frac{5}{4}$	$\frac{5}{4}$	1
$g_2^2(y_2)_{ij}$	$-\frac{9}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-\frac{33}{4}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{3}{4}$	$-\frac{15}{4}$	$-\frac{3}{2}$
$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	$\frac{3}{2}$	0	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	$\frac{1}{2}$	0	0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$T_\nu^{(2)}(y_2)_{ij}$	2	2	1	1	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	1	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0	0	0	0	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0	0	0	0	0

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$T_\nu^{(2)}(y_2)_{ij}$	1
$(y_1(y_1)^\dagger y_2)_{ij}$	$\frac{3}{4}$
$(Y_e(Y_e)^\dagger y_2)_{ij}$	$\frac{1}{2}$
$(y_2(y_2)^\dagger y_2)_{ij}$	$\frac{3}{2}$
$g_2^2(y_2)_{ij}$	$-\frac{9}{2}$
$g_1^2(y_2)_{ij}$	$-\frac{15}{2}$
$\text{Tr}(y_2(y_1)^\dagger)(y_1)_{ij}$	0
$(y_2(y_1)^\dagger y_1)_{ij}$	0

β_{M_F}

Run block 1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$(y_1^T y_1^*) M_F$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	1
$M_F(y_2^\dagger y_2)$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	1	1
$g_2^2 M_F$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	0	0

$g_1^2 M_F$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	-6	0
-------------	-----------------	----------------	----------------	------	-----

Run block 2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$(y_1^T y_1^*) M_F$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$
$g_2^2 M_F$	0	-12	-12	-12	$-\frac{9}{2}$
$g_1^2 M_F$	-6	-6	0	-6	$-\frac{3}{2}$

Run block 3

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$(y_1^T y_1^*) M_F$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_2^2 M_F$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 M_F$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$

Run block 4

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$(y_1^T y_1^*) M_F$	$\frac{3}{4}$
$M_F(y_2^\dagger y_2)$	$\frac{3}{4}$
$g_2^2 M_F$	$-\frac{9}{2}$
$g_1^2 M_F$	$-\frac{27}{2}$

$\beta_{m_1^2}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$m_1^2 T_\nu^{(1)}$	4	4	4	2	2
$M_F^\dagger M_F T_\nu^{(1)}$	-8	-8	-8	-4	-16
$\lambda_{S_1}^{(1)} m_1^2$	4	4	4	6	6
$g_2^2 m_1^2$	0	0	0	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_1^2$	-24	-6	0	$-\frac{27}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_2^2$	6	6	6	4	4

$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$m_1^2 T_\nu^{(1)}$	2	3	3	3	2
$M_F^\dagger M_F T_\nu^{(1)}$	-4	-6	-24	-6	-4
$\lambda_{S_1}^{(1)} m_1^2$	6	6	6	6	8
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-12
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	-6
$\lambda_{12}^{(1)} m_2^2$	4	4	4	4	2
$\lambda_{S_1}^{(A)} m_1^2$	0	0	0	0	4
$\lambda_{12}^{(\times)} m_2^2$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$m_1^2 T_\nu^{(1)}$	2	2	2	2	2
$M_F^\dagger M_F T_\nu^{(1)}$	-4	-4	-4	-4	-4
$\lambda_{S_1}^{(1)} m_1^2$	8	8	8	8	8
$g_2^2 m_1^2$	-12	-12	-12	-12	-12
$g_1^2 m_1^2$	0	-6	-24	-6	0
$\lambda_{12}^{(1)} m_2^2$	2	2	6	6	6
$\lambda_{S_1}^{(A)} m_1^2$	4	4	4	4	4
$\lambda_{12}^{(\times)} m_2^2$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$m_1^2 T_\nu^{(1)}$	2
$M_F^\dagger M_F T_\nu^{(1)}$	-4
$\lambda_{S_1}^{(1)} m_1^2$	8
$g_2^2 m_1^2$	-12
$g_1^2 m_1^2$	-6
$\lambda_{12}^{(1)} m_2^2$	6
$\lambda_{S_1}^{(A)} m_1^2$	4

$\lambda_{12}^{(\times)} m_2^2$	4
---------------------------------	---

$\beta_{m_2^2}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$m_2^2 T_\nu^{(2)}$	2	2	2	2	2
$M_F^\dagger M_F T_\nu^{(2)}$	-4	-4	-4	-4	-16
$\lambda_{S_2}^{(1)} m_2^2$	8	8	8	6	6
$g_2^2 m_2^2$	-12	-12	-12	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	-6	0	-6	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_1^2$	2	2	2	4	4
$\lambda_{S_2}^{(A)} m_2^2$	4	4	4	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$m_2^2 T_\nu^{(2)}$	2	3	3	3	4
$M_F^\dagger M_F T_\nu^{(2)}$	-4	-6	-24	-6	-8
$\lambda_{S_2}^{(1)} m_2^2$	6	6	6	6	4
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	0
$g_1^2 m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$	0
$\lambda_{12}^{(1)} m_1^2$	4	4	4	4	6
$\lambda_{S_2}^{(A)} m_2^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} m_1^2$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$m_2^2 T_\nu^{(2)}$	4	4	2	2	2
$M_F^\dagger M_F T_\nu^{(2)}$	-8	-8	-4	-4	-4
$\lambda_{S_2}^{(1)} m_2^2$	4	4	8	8	8
$g_2^2 m_2^2$	0	0	-12	-12	-12
$g_1^2 m_2^2$	-6	-24	-6	0	-6
$\lambda_{12}^{(1)} m_1^2$	6	6	6	6	6

$\lambda_{S_2}^{(A)} m_2^2$	0	0	4	4	4
$\lambda_{12}^{(\times)} m_1^2$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$m_2^2 T_\nu^{(2)}$	2
$M_F^\dagger M_F T_\nu^{(2)}$	−4
$\lambda_{S_2}^{(1)} m_2^2$	8
$g_2^2 m_2^2$	−12
$g_1^2 m_2^2$	−24
$\lambda_{12}^{(1)} m_1^2$	6
$\lambda_{S_2}^{(A)} m_2^2$	4
$\lambda_{12}^{(\times)} m_1^2$	4

$\beta_{\lambda_{HS_1}^{(1)}}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$\lambda_{HS_1}^{(1)} T_\nu^{(1)}$	4	4	4	2	2
$\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	−4	−4	−4	−2	−2
$ \lambda_5 ^2$	12	12	12	6	6
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	−9	−9
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	−15	−3
$\lambda_1 \lambda_{HS_1}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	4	4	4	6	6
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6	6	6	4	4
g_2^4	0	0	0	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	12	3	0	$\frac{27}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$\lambda_{HS_1}^{(1)} T_\nu^{(1)}$	2	3	3	3	2
$\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2	-3	-3	-3	-2
$ \lambda_5 ^2$	6	6	6	6	4
$g_2^2 \lambda_{HS_1}^{(1)}$	-9	-9	-9	-9	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	-3	-15	-3	-3	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)2}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	2
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	6
g_1^4	$\frac{3}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	3
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$\lambda_{HS_1}^{(1)} T_\nu^{(1)}$	2	2	2	2	2
$\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	-2	-2	-2	-2	-2
$ \lambda_5 ^2$	4	4	4	4	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)2}$	6	6	6	6	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	2	2	6	6	6
g_2^4	6	6	6	6	6
g_1^4	0	3	12	3	0
$\lambda_{HS_1}^{(1)} T$	2	2	2	2	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$\lambda_{HS_1}^{(1)} T_\nu^{(1)}$	2
$\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$	−2
$ \lambda_5 ^2$	4
$g_2^2 \lambda_{HS_1}^{(1)}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_1}^{(1)}$	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_1}^{(1)}$	6
$\left(\lambda_{HS_1}^{(1)}\right)^2$	4
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$	6
g_2^4	6
g_1^4	3
$\lambda_{HS_1}^{(1)} T$	2
$\left(\lambda_{HS_1}^{(A)}\right)^2$	2
$\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(A)}$	4
$\lambda_{HS_2}^{(1)} \lambda_{12}^{(\times)}$	4

$\beta_{\lambda_{HS_2}^{(1)}}$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$\lambda_{HS_2}^{(1)} T_\nu^{(2)}$	2	2	2	2	2
$\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	−2	−2	−2	−2	−2
$ \lambda_5 ^2$	4	4	4	6	6
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	−9	−9
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$	−3	−3
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	6	6
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	2	2	2	4	4
g_2^4	6	6	6	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	3	0	3	$\frac{3}{4}$	$\frac{3}{4}$
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2	2	2	$\frac{3}{4}$	$\frac{3}{4}$

$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	0	0
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$\lambda_{HS_2}^{(1)} T_\nu^{(2)}$	2	3	3	3	4
$\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-2	-3	-3	-3	-4
$ \lambda_5 ^2$	6	6	6	6	12
$g_2^2 \lambda_{HS_2}^{(1)}$	-9	-9	-9	-9	$-\frac{9}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	-15	-3	-3	-15	$-\frac{3}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	4
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	4	4	4	4	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	0
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$	0
$\left(\lambda_{HS_2}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	0
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$\lambda_{HS_2}^{(1)} T_\nu^{(2)}$	4	4	2	2	2
$\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	-4	-4	-2	-2	-2
$ \lambda_5 ^2$	12	12	4	4	4
$g_2^2 \lambda_{HS_2}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6	6	6	6	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	4	4	8	8	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6	6	6	6	6
g_2^4	0	0	6	6	6

g_1^4	3	12	3	0	3
$\left(\lambda_{HS_2}^{(A)}\right)^2$	0	0	2	2	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	0	0	4	4	4
$\lambda_{HS_2}^{(1)} T$	2	2	2	2	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$\lambda_{HS_2}^{(1)} T_\nu^{(2)}$	2
$\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$	−2
$ \lambda_5 ^2$	4
$g_2^2 \lambda_{HS_2}^{(1)}$	− $\frac{33}{2}$
$g_1^2 \lambda_{HS_2}^{(1)}$	− $\frac{51}{2}$
$\lambda_1 \lambda_{HS_2}^{(1)}$	6
$\left(\lambda_{HS_2}^{(1)}\right)^2$	4
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$	8
$\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$	6
g_2^4	6
g_1^4	12
$\left(\lambda_{HS_2}^{(A)}\right)^2$	2
$\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(A)}$	4
$\lambda_{HS_2}^{(1)} T$	2
$\lambda_{HS_1}^{(1)} \lambda_{12}^{(\times)}$	4

$$\beta_{\lambda_{12}^{(1)}}$$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$\lambda_{12}^{(1)} T_\nu^{(1)}$	4	4	4	0	0
$\lambda_{12}^{(1)} T_\nu^{(2)}$	2	2	2	0	0
$\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	−4	−4	−4	−2	−2
$ \lambda_5 ^2$	4	4	4	3	3
$g_2^2 \lambda_{12}^{(1)}$	−12	−12	−12	−9	−9
$g_1^2 \lambda_{12}^{(1)}$	−30	−6	−6	−15	−3

$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	4	4	4	6	6
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8	8	8	6	6
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	0	0	0	$\frac{9}{4}$	$\frac{9}{4}$
g_1^4	48	0	0	$\frac{27}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4	4	4	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{12}^{(1)} T_\nu$	0	0	0	2	2
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$\lambda_{12}^{(1)} T_\nu^{(1)}$	0	0	0	0	2
$\lambda_{12}^{(1)} T_\nu^{(2)}$	0	0	0	0	4
$\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	-2	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-4
$ \lambda_5 ^2$	3	3	3	3	4
$g_2^2 \lambda_{12}^{(1)}$	-9	-9	-9	-9	-12
$g_1^2 \lambda_{12}^{(1)}$	-15	-15	-3	-15	-6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	6	6	6	6	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	6	6	6	6	4
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$	0
g_1^4	$\frac{27}{4}$	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$	0
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$	0
$\lambda_{12}^{(1)} T_\nu$	2	3	3	3	0
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	0	0	0	0	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0

$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0
---	---	---	---	---	---

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$\lambda_{12}^{(1)} T_\nu^{(1)}$	2	2	0	0	0
$\lambda_{12}^{(1)} T_\nu^{(2)}$	4	4	0	0	0
$\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	−4	−4	−2	−2	−2
$ \lambda_5 ^2$	4	4	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	−12	−12	−24	−24	−24
$g_1^2 \lambda_{12}^{(1)}$	−6	−30	−30	−6	−6
$\left(\lambda_{12}^{(1)}\right)^2$	4	4	4	4	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8	8	8	8	8
$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	4	4	8	8	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4	4	4	4	4
g_2^4	0	0	12	12	12
g_1^4	0	48	48	0	0
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	0	0	4	4	4
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	4	4	4
$\lambda_{12}^{(1)} T_\nu^{(1)}$	0	0	2	2	2
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4	4	4	4	4
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	4	4	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	4	4	4
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	4	4	4

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$\lambda_{12}^{(1)} T_\nu^{(1)}$	0
$\lambda_{12}^{(1)} T_\nu^{(2)}$	0
$\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$	−2
$ \lambda_5 ^2$	$\frac{4}{3}$
$g_2^2 \lambda_{12}^{(1)}$	−24
$g_1^2 \lambda_{12}^{(1)}$	−30
$\left(\lambda_{12}^{(1)}\right)^2$	4
$\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$	8

$\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$	8
$\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$	4
g_2^4	12
g_1^4	48
$\lambda_{12}^{(1)} \lambda_{S_2}^{(A)}$	4
$\left(\lambda_{12}^{(A)}\right)_2$	4
$\lambda_{12}^{(1)} T_\nu$	2
$\lambda_{12}^{(1)} \lambda_{S_1}^{(A)}$	4
$\left(\lambda_{12}^{(\times)}\right)_2$	4
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	4
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	4

β_{λ_5}

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$	-9	-9
$g_1^2 \lambda_5$	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	-9	-3
$\lambda_5 T_\nu^{(1)}$	2	2	2	0	0
$\lambda_5 T_\nu^{(2)}$	1	1	1	0	0
$\lambda_5 \lambda_{HS_2}^{(A)}$	4	4	4	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	0	0	0	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	0	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$
$\lambda_5 T_\nu$	0	0	0	1	1
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4

$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	-9	-9	-9	-9	$-\frac{21}{2}$
$g_1^2 \lambda_5$	-9	-9	-3	-9	$-\frac{9}{2}$
$\lambda_5 T_\nu^{(1)}$	0	0	0	0	1
$\lambda_5 T_\nu^{(2)}$	0	0	0	0	2
$\lambda_5 \lambda_{HS_2}^{(A)}$	2	2	2	2	0
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2	-2	-2	-2	-4
$\lambda_5 \lambda_{12}^{(A)}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0
$\lambda_5 T_\nu$	1	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$	0
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	0	0	0

Run block 3: $T3\text{-}D\text{-}p0\text{-}T3\text{-}E\text{-}p0$

$\mathcal{S}$	T3 - D - p0 ($\alpha = 0$)	T3 - D - p2 ($\alpha = 2$)	T3 - E - m4 ($\alpha = -4$)	T3 - E - m2 ($\alpha = -2$)	T3 - E - p0 ($\alpha = 0$)
$\lambda_5 \lambda_1$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4	4	4	4	4
$\lambda_5 \lambda_{12}^{(1)}$	2	2	2	2	2
$g_2^2 \lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lambda_5 T_\nu^{(1)}$	1	1	0	0	0
$\lambda_5 T_\nu^{(2)}$	2	2	0	0	0
$\lambda_5 \lambda_{HS_2}^{(A)}$	0	0	2	2	2
$\lambda_5 T$	2	2	2	2	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-4	-4	-2	-2	-2
$\lambda_5 \lambda_{12}^{(A)}$	0	0	2	2	2
$\lambda_5 T_\nu$	0	0	1	1	1
$\lambda_5 \lambda_{12}^{(\times)}$	0	0	-2	-2	-2

Run block 4: $T3\text{-}E\text{-}p2\text{-}T3\text{-}E\text{-}p2$

$\mathcal{S}$	T3 - E - p2 ($\alpha = 2$)
$\lambda_5 \lambda_1$	2
$\lambda_5 \lambda_{HS_1}^{(1)}$	4
$\lambda_5 \lambda_{HS_2}^{(1)}$	4
$\lambda_5 \lambda_{12}^{(1)}$	2

$g_2^2 \lambda_5$	$-\frac{33}{2}$
$g_1^2 \lambda_5$	$-\frac{33}{2}$
$\lambda_5 T_\nu^{(1)}$	0
$\lambda_5 T_\nu^{(2)}$	0
$\lambda_5 \lambda_{HS_2}^{(A)}$	2
$\lambda_5 T$	2
$\lambda_5 \lambda_{HS_1}^{(A)}$	-2
$\lambda_5 \lambda_{12}^{(A)}$	2
$\lambda_5 T_\nu$	1
$\lambda_5 \lambda_{12}^{(\times)}$	-2

Additional scalar-coupling beta functions

$$\beta_{\lambda_{HS_1}^{(A)}}$$

Run block 1: $T3\text{-}A\text{-}m4\text{-}T3\text{-}B\text{-}m1$

$\mathcal{S}$	T3 - A - m4 ($\alpha = -4$)	T3 - A - m2 ($\alpha = -2$)	T3 - A - p0 ($\alpha = 0$)	T3 - B - m3 ($\alpha = -3$)	T3 - B - m1 ($\alpha = -1$)
$g_1^2 g_2^2$	0	0	0	-18	-6

Run block 2: $T3\text{-}B\text{-}p1\text{-}T3\text{-}D\text{-}m2$

$\mathcal{S}$	T3 - B - p1 ($\alpha = 1$)	T3 - C - m3 ($\alpha = -3$)	T3 - C - m1 ($\alpha = -1$)	T3 - C - p1 ($\alpha = 1$)	T3 - D - m2 ($\alpha = -2$)
$g_1^2 g_2^2$	6	-18	-6	6	-12

Run block 3: $T3\text{-}D\text{-}p0\text{-}T3\text{-}E\text{-}p0$

$\mathcal{S}$	T3 - D - p0 ($\alpha = 0$)	T3 - D - p2 ($\alpha = 2$)	T3 - E - m4 ($\alpha = -4$)	T3 - E - m2 ($\alpha = -2$)	T3 - E - p0 ($\alpha = 0$)
$g_1^2 g_2^2$	0	12	-24	-12	0

Run block 4: $T3\text{-}E\text{-}p2\text{-}T3\text{-}E\text{-}p2$

$\mathcal{S}$	T3 - E - p2 ($\alpha = 2$)
$g_1^2 g_2^2$	12

$$\beta_{\lambda_{HS_2}^{(A)}}$$

Run block 1: $T3-A-m4-T3-B-m1$

$\mathcal{S}$	$T3-A-m4 (\alpha = -4)$	$T3-A-m2 (\alpha = -2)$	$T3-A-p0 (\alpha = 0)$	$T3-B-m3 (\alpha = -3)$	$T3-B-m1 (\alpha = -1)$
$g_1^2 g_2^2$	-12	0	12	-6	6

Run block 2: $T3-B-p1-T3-D-m2$

$\mathcal{S}$	$T3-B-p1 (\alpha = 1)$	$T3-C-m3 (\alpha = -3)$	$T3-C-m1 (\alpha = -1)$	$T3-C-p1 (\alpha = 1)$	$T3-D-m2 (\alpha = -2)$
$g_1^2 g_2^2$	18	-6	6	18	0

Run block 3: $T3-D-p0-T3-E-p0$

$\mathcal{S}$	$T3-D-p0 (\alpha = 0)$	$T3-D-p2 (\alpha = 2)$	$T3-E-m4 (\alpha = -4)$	$T3-E-m2 (\alpha = -2)$	$T3-E-p0 (\alpha = 0)$
$g_1^2 g_2^2$	0	0	-12	0	12

Run block 4: $T3-E-p2-T3-E-p2$

$\mathcal{S}$	$T3-E-p2 (\alpha = 2)$
$g_1^2 g_2^2$	24

$$\beta_{\lambda_{12}^{(A)}}$$

Run block 1: $T3-A-m4-T3-B-m1$

$\mathcal{S}$	$T3-A-m4 (\alpha = -4)$	$T3-A-m2 (\alpha = -2)$	$T3-A-p0 (\alpha = 0)$	$T3-B-m3 (\alpha = -3)$	$T3-B-m1 (\alpha = -1)$
$g_1^2 g_2^2$	0	0	0	18	-6

Run block 2: $T3-B-p1-T3-D-m2$

$\mathcal{S}$	$T3-B-p1 (\alpha = 1)$	$T3-C-m3 (\alpha = -3)$	$T3-C-m1 (\alpha = -1)$	$T3-C-p1 (\alpha = 1)$	$T3-D-m2 (\alpha = -2)$
$g_1^2 g_2^2$	18	18	-6	18	0

Run block 3: $T3-D-p0-T3-E-p0$

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
$g_1^2 g_2^2$	0	0	48	0	0

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
$g_1^2 g_2^2$	48

$$\beta_{\lambda_{12}^{(\times)}}$$

Run block 1: T3-A-m4-T3-B-m1

$\mathcal{S}$	T3 – A – m4 ($\alpha = -4$)	T3 – A – m2 ($\alpha = -2$)	T3 – A – p0 ($\alpha = 0$)	T3 – B – m3 ($\alpha = -3$)	T3 – B – m1 ($\alpha = -1$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 2: T3-B-p1-T3-D-m2

$\mathcal{S}$	T3 – B – p1 ($\alpha = 1$)	T3 – C – m3 ($\alpha = -3$)	T3 – C – m1 ($\alpha = -1$)	T3 – C – p1 ($\alpha = 1$)	T3 – D – m2 ($\alpha = -2$)
g_2^4	0	0	0	0	0
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	0	0	0
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	0	0	0
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	0	0	0

Run block 3: T3-D-p0-T3-E-p0

$\mathcal{S}$	T3 – D – p0 ($\alpha = 0$)	T3 – D – p2 ($\alpha = 2$)	T3 – E – m4 ($\alpha = -4$)	T3 – E – m2 ($\alpha = -2$)	T3 – E – p0 ($\alpha = 0$)
---------------	------------------------------	------------------------------	-------------------------------	-------------------------------	------------------------------

g_2^4	0	0	6	6	6
$\left(\lambda_{12}^{(A)}\right)^2$	0	0	2	2	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	0	0	8	8	8
$\left(\lambda_{12}^{(\times)}\right)^2$	0	0	10	10	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	0	0	−2	−2	−2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	0	0	−2	−2	−2

Run block 4: T3-E-p2-T3-E-p2

$\mathcal{S}$	T3 – E – p2 ($\alpha = 2$)
g_2^4	6
$\left(\lambda_{12}^{(A)}\right)^2$	2
$\lambda_{12}^{(1)} \lambda_{12}^{(\times)}$	8
$\left(\lambda_{12}^{(\times)}\right)^2$	10
$\lambda_{12}^{(\times)} \lambda_{S_1}^{(A)}$	−2
$\lambda_{12}^{(\times)} \lambda_{S_2}^{(A)}$	−2

First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether F , S_1 , S_2 , or a degenerate set is integrated out first.